

1. Design & implement Role-based access control (RBAC) to regulate access permissions in an online banking system ensuring employees receive privileges according to their roles.

Ans) RBAC in online Banking system:-

Design:-
Role based access control (RBAC) assigns permissions based on roles instead of individuals.

Roles & permissions:-

Customer → views balance, transfer money.
Teller → process deposits / withdrawals.
Manager → Approve large transactions, view reports.
Admin → Manage users & system settings

Implementation:-

- Create roles in the system.
- Assign permissions to each role.
- Assign users (employees) to roles.
- Use authentication (login) & authorization (role check).

Benefit:-

- Ensures employee only access what they need → improves security & reduces misuse.

2. Examine the resource allocation approach of an organization running multiple cloud applications, identify underutilized and overutilized resources and assess how this imbalance influences overall cloud expenditure.

Ans) Cloud Resource Allocation Analysis:-

Approach:-

• Monitor usage of CPU, memory, storage.

Identify:-

Underutilized resources → Low usage (wasting money)

Overutilized resources → High usage (performance issues)

Impact on Cost:-

Underutilized → unnecessary cloud expenses.

Overutilized → System slowdown, possible downtime.

Solution:-

Auto-scaling (adjust resources automatically)

Right sizing (match resources to workload)

Load Balancing.

Conclusion:-

Balanced allocation reduces cost and improves performance.

3. Identify how Identity and Access Management (IAM) can be implemented to assign suitable access permissions to students, faculty, administrations in a university cloud system handling both academic & administrative data.

Ans) IAM in university cloud system

Roles:-

Students → Access courses, results

Faculty → Upload materials, grade systems

Admin → Manage users, system data

Implementation:-

- Create IAM roles for each group
- Assign (~~responsibilities~~) permissions based on responsibilities
- Use authentication (ID/password, MFA)
- Apply least privilege principle.

Example:- a student cannot modify grades but faculty can.

Benefits:- Protects both academic and administrative data security.

Q. Access the identity and Access Management (IAM) setup of a multinational organisation, identify cases from employees have either restricted or excessive access and recommend measures to achieve balanced and secured access control.

Ans) IAM Assessment in Multinational Organisation

Issues Identified:-

- 1) Excessive access → employees have more permissions than needed (security risk)
- 2) Restricted access → Employees cannot perform tasks (reduces productivity)

Solutions:-

- > Apply least privilege principle
- > Regular access audits
- > Use role-based access control (RBAC)
- > Implement Multi Factor authentication
- > Remove unused accounts.

Outcome:- Balanced access improves both security and efficiency.

5. Illustrate how client-based access can support secure connectivity for employees accessing cloud applications through their laptops.

Ans) Client-Based Access for secure connectivity

Concept:- Client-Based access allows employees to securely connect to cloud systems using their laptops.

• Implementation:-

- > Install secure client software (VPN or secure apps)
- > Use authentication (username, password > MFA)
- > Encrypt data during transmissions
- > Apply device security checks (antivirus, updates)

Example:-

> Employee logs in via VPN to access company cloud securely from home.

• Benefits:-

- > Secure remote access
- > Protects sensitive data
- > Prevents unauthorized access —

10